

## **SECTION 15620-A CENTRIFUGAL CHILLER**

### PART 1 - GENERAL

#### 1.1 DESCRIPTION

- A. This Section specifies the requirements for provision and installation of main water chillers. Other work associated with the chillers, including piping, insulation and controls, is covered in other Section.
- B. This Section specifies includes performance, proprietary.

#### 1.2 WORK INCLUDED

- A. This Section specifies the requirements necessary to furnish and install, complete, water-cooled centrifugal chillers.

#### 1.3 REFERENCE CODES

- A. The Contractor shall submit certification that the chiller supplied conforms to the ANSI, ASHRAE, ASME, AHRI, UL and NEMA standard.
- B. Chinese National Standards shall govern when available and where applicable. The Contractor may propose use of internationally recognized codes and standards. Acceptance of such alternates shall be subject to approval by the Client.
- C. Any conflict between this specification and the requirements of the listed references or between this specification and its data sheets or drawings shall be brought immediately to the attention of the Client for resolution.

#### 1.4 QUALITY ASSURANCE

Rate chillers in accordance with AHRI standard 550/590. Machine construction and safety devices shall conform with ANSI B9.1.

#### 1.5 SUBMITTALS

The Contractor shall complete and submit six (6) copies of the following:

A. Vendor Documents

Submit shop drawings and manufacturer's data. These drawings shall show equipment plans, elevations, and sections, dimension of equipment, electrical wiring diagram, controls, piping connection arrangements and floor loading requirements. The manufacturer's data shall indicate total electrical loads, compressor motor rating, electrical characteristics, sequence of operation, and installation and testing procedures.

B. Maintenance Data and Operating Instructions

Submit manufacturer's printed operation and maintenance instructions, and describe the operating procedures and maintenance sequences. The Contractor shall also include the manufacturer's recommended spare parts list and maintenance instructions.

C. Tests

Submit reports of factory and certificates of compliance and show the results of operating tests.

D. Submit product data indicating rated capacities, weights, specialties and accessories, electrical requirements and wiring diagrams.

E. Submit written certification that components of package not furnished by manufacturer have been selected in accordance with manufacturer's requirements.

F. Submit manufacturer installation instruction.

G. Submit performance data indicating energy input versus cooling load output from 25 to 100 percent of full load at specified and minimum condenser water temperature.

## PART 2 - PRODUCTS

### 2.1 GENERAL

Each chiller shall be a complete factory package comprising centrifugal Dual compressors, open or hermetic motors, cooler, evaporator, condenser, refrigerant flow control device(s), refrigerant storage receiver, motor starter, and control center with related unit mounted-controls. The unit shall be factory assembled, piped, wired, leak tested, evacuated and partially charged with refrigerant prior to dispatch from the factory. Refrigerant must be low toxicity, non-flammable and GWP lower than 150 to meet F-gas regulation, R-1233zd and R-514A, or other refrigerants accepted by owner.

### 2.2 BASIC DATA

- A. Quantity: SIX (6) Set -12°C\*6 Sets
- B. Type: Centrifugal
- C. Capacity: 12°C-2200 RT
- D. Multiple flue TWO pass
- E. Evaporator temperature : Outlet 12°C / Inlet 20°C
- F. Evaporator flow rate: 231 L/S
- G. Condenser temperature : Outlet 38.0°C / Inlet 32.0°C
- H. Condenser flow rate: 356 L/S
- I. Electrical : 6600V, 3-phase, 60Hz (HP)

### 2.3 COMPRESSORS - MOTOR A.

#### Guide Vanes

Fully modulating variable inlet guide vanes provide capacity control. The guide vanes are controlled by internal oil pressure driven or externally-mounted electric vane operator in response to refrigeration load on the evaporator.

#### B. Impellers

Fully shrouded impellers made of high strength aluminum

alloy are directly connected to the motor rotor shaft operating at 3,600 rpm (60 Hz). The impellers are dynamically balanced and over-speed tested at 4,500 rpm; the motor-compressor assembly is balanced to a maximum vibration of 0.15 inch/second at 3,600 rpm as measured on the motor housing.

#### C. Compressor Casing

Separate volute casings of refrigerant-tight, close-grained cast iron are used on the centrifugal compressor; each incorporating a parallel wall diffuser surrounded by a collection scroll. The diffuser passages are machined to ensure high efficiency. All casings are proof- and leak-tested.

#### D. Motor

Compressor motors are hermetically sealed two-pole, squirrel cage induction-type. They are built in accordance with Trane specifications and guaranteed by the manufacturer for continuous operation at the nameplate rating. A load-limit system provides protection against operation in excess of this rating. The rotor shaft is heat-treated carbon steel and designed such that the critical speed is well above the operating speed. The control circuit prevents motor energization unless positive oil pressure is established. Impellers are keyed directly to the motor shaft and locked in position. Nonferrous, labyrinth-type seals minimize recirculation and gas leakage between the stages of the compressor. 4160V, 3-phase 60 Hz motors are supplied with three terminal posts for reduced-voltage (primary reactor) starting. Motor terminal pads are supplied. A removable sheet metal terminal box encloses the terminal board area.

#### E. Motor Cooling

Motor cooling is accomplished by a patented refrigerant pump that supplies liquid refrigerant to the motor. The refrigerant circulates uniformly over the stator windings and between the rotor and stator. All motor windings are specifically insulated for operation within a refrigerant atmosphere. F.

#### Lubrication

The oil pump for low voltage Series E chillers is driven by a 380-600V, 50/60 Hz, 3-phase, 2 hp motor, while the oil pump for medium voltage models is driven

by a 200-240V, 50/60 Hz, 1-phase, 2 hp motor. The motor and pump assembly are submerged in the oil sump to ensure a positive oil supply to the compressor bearings at all times. A low watt density heater maintains the oil temperature to minimize its affinity for refrigerant.

The oil tank is constructed in accordance with ASME Section VIII, Division I. It is designed with an open internal volume to accommodate the separation of refrigerant vapor from oil during operation. An electrically actuated ball valve prevents foaming and oil loss during a chiller start. It utilizes two heater elements and a cooling sub-system that consists of a small brazed plate heat exchanger working in combination with a solenoid valve.

## 2.4 EVAPORATOR

### A. Shell and Waterboxes

The evaporator shell is constructed of carbon steel plate and incorporates a steel rupture disc in accordance with the ASME Section VIII, Division I. A refrigerant temperature coupling is provided for a low limit controller or customer use.

Multiple pass arrangements are available at 1040 kPa (150 psig) water side working pressures, with grooved connections. Flanged connections waterboxes are also available.

### B. Tube Sheets

A thick carbon steel tube sheet is welded to each end of the shell and is drilled and reamed to accommodate the tubes. Three annular grooves are machined into each tube hole to provide a positive liquid and vapor seal between the refrigerant and water side of the shell after tube rolling. Intermediate tube support sheets are positioned along the length of the shell to avoid contact and relative motion between adjacent tubes.

### C. Tubes

Individually replaceable, seamless tubing available in a variety of materials, depending on the customer's needs, is used as the evaporator heat transfer surface; tubing is available in either 19 mm (0.75 inch) outside diameter. Tubes are externally and internally enhanced, and mechanically expanded into the tube sheets (and are secured to the intermediate supports with tube clips) to provide a leak-free seal and eliminate tube contact and abrasion due to relative motion.

#### D. Eliminators

Multiple layers of metal mesh screen form the eliminators and are installed over the tube bundle along the entire length of the evaporator. The eliminators prevent liquid refrigerant carryover into the compressor.

#### E. Refrigerant Distribution

A refrigerant distributor on the base of the evaporator assures uniform wetting of the heat transfer surface over the entire length of the shell and under varying loads. High velocity, refrigerant-spray impingement on the tubes is prevented through this design.

#### F. Refrigerant Flow Control

A multiple orifice flow-control system maintains the correct pressure differential between the condenser, economizer, and evaporator over the entire range of loading. This patented system contains no moving parts.

#### G. Shell Tests

The refrigerant side of the evaporator shell, complete with tubes but without waterbox covers, is proof-tested at 390kPa (55Psi) for ASME, vacuum leak-tested, and finally pressure leak-tested with a helium mass spectrometer. The water side of the evaporator shell, with waterboxes in place, is hydrostatically tested at 1.5 times the design pressure for ASME, 1560kPa (225Psi) for 1040kPa (150Psi) waterboxes; 1.25 times the design pressure for ASME, 1300kPa (188Psi) for 1040kPa (150Psi) waterboxes.

### 2.5 CONDENSER

#### A. Shell and Waterboxes

The condenser shell is constructed of carbon steel plate designed and constructed in accordance with ASME Section VIII, Division I. Multiple pass arrangements are available at 1040Kpa (150 psig) water side working pressures, with grooved connections. Flanged connections and/or marine-type waterboxes are also available.

#### B. Tube Sheets

A thick carbon steel tube sheet is welded to each end

of the shell and is drilled and reamed to accommodate the tubes. Three annular grooves are machined into each tube hole to provide a positive liquid and vapor seal between the refrigerant and water sides of the shell after tube rolling. Intermediate tube support sheets are positioned along the length of the shell to avoid contact and relative motion between adjacent tubes.

#### C. Tubes

Individually replaceable, seamless copper tubing available in either 25 mm (1 inch) or 19 mm (0.75 inch) outside diameter is used as the evaporator heat transfer surface. Tubes are externally and internally enhanced, and mechanically expanded into the tube sheets (and are secured to the intermediate supports with tube clips) to provide a leak-free seal and eliminate tube contact and abrasion due to relative motion.

#### D. Refrigerant Gas Distribution

A baffle plate between the tube bundle and the condenser shell distributes the hot compressor-discharge gas longitudinally throughout the condenser and downward over the tube bundle. The baffle plate prevents direct impingement of high velocity compressor-discharge gas upon the tubes.

#### E. Shell Tests

The refrigerant side of the condenser shell, complete with tubes but without waterbox covers, is proof-tested at 390kPa (55Psi) for ASME, vacuum leak-tested, and finally pressure leak-tested with a helium mass spectrometer. The water side of the evaporator shell, with waterboxes in place, is hydrostatically tested at 1.5 times the design pressure for ASME, 1560kPa (225Psi) for 1040kPa (150Psi) waterboxes; 1.25 times the design pressure for ASME, 1300kPa (188Psi) for 1040kPa (150Psi) waterboxes.

### 2.6 ECONOMIZER

The 2-stage (60 Hz) chiller utilizes a single stage economizer. The economizer is constructed in accordance with ASME Section VIII, Division I and consists of either one or two interstage pressure chambers which utilize a multiple orifice system to maintain the correct pressure differential between the condenser, economizer, and evaporator over the entire range of loading. This patented system contains no moving parts.

## 2.7 PURGE SYSTEM

### Standard Features

- 115Vac, 50/60 Hz, 1-Phase.
- 175 watt carbon tank heater.
- 12.3 minimum circuit ampacity.
- 2411.1 kPa (335 psig) design pressure high side.
- 1307.9 kPa (175 psig) design pressure low side.
- The purge is 654 mm (25.75 in.) high, 699 mm (27.5 in.) wide, and 552 mm (21.75 in.) deep.
- The purge uses an R-404A refrigeration circuit with a 1/4-hp condensing unit/10.3 total unit amps (fan, compressor, expansion valve), and a compressor suction temperature sensor.

The purge tank has a fusible plug, evaporator coil, normally-closed float switch, and the following connections:

- 6 mm (1/4 in.) liquid return with filter-drier and moisture indicator
- 16 mm (5/8 in.) vapor line

The expansion valve automatically controls the purge suction pressure to 234.4 kPa (34 psia). The pump-out system consists of a pump-out compressor, pump-out solenoid valve, and an exhaust solenoid valve.

The carbon bed tank incorporates a temperature sensor and a regenerative cycle, a 175-watt resistive heater, 1034.2 kPa (150 psi) pressure relief valve, and a temperature sensor. The carbon bed tank automatically collects and scrubs refrigerant molecules from the noncondensable gas and drives any collected refrigerant vapor back into the chiller. This design keeps the purge efficiency at peak levels throughout its life without the maintenance required on other purges.

The purge controller interfaces with the following intelligent devices on an IPC3 communications link: liquid-level switch, dual relay output, quad relay output, dual triac output, suction temperature sensor, and carbon temperature sensor. Fifty hertz applications have a separate voltage correction transformer.

The purge controller communicates with the chiller controller and display, which is mounted on the front of the chiller control panel. Descriptive text indicates purge operating mode, status, set points, purge operating data reports, diagnostics, and alarms. Operating modes Stop, On, Auto, and Adaptive operate the purge



refrigeration circuit and accumulate noncondensables with or without the chiller running.

## 2.8 CHILLER CONTROLLER

The microcomputer control panel is factory installed and tested on the chiller. All controls necessary for the safe and reliable operation of the chiller are provided, purge operation, and interface to the starter. The control system is powered by a control power transformer included in the starter panel. The microcomputer control system processes the leaving evaporator fluid temperature sensor signal to satisfy the system requirements across the entire load range.

The microprocessor controller is compatible with reduced-voltage electromechanical starters. Depending on the applicability, the drives may be factory mounted or remote mounted.

The controller will load and unload the chiller via control of the stepper motor/actuator which drives the inlet guide vanes open or closed. The load range can be limited either by a current limiter or by an inlet guide vane limit (whichever controls the lower limit). It will also control the evaporator and condenser pumps to insure proper chiller operation.

Approximately 200 diagnostic checks are made and displayed when a fault is detected. The display indicates the fault, the type of reset required, the time and date the diagnostic occurred, the mode in which the machine was operating at the time of the diagnostic, and a help message. A diagnostic history displays the last 10 diagnostics with the time and date of their occurrence.

The panel features machine protection shutdown requiring manual reset for:

- ☐ Low oil temperature
- ☐ Actuator drive circuit fault
- ☐ Low differential oil pressure
- ☐ Extended compressor surge
- ☐ Excessive loss of communication
- ☐ High condenser refrigerant pressure
- ☐ Critical sensor or detection circuit faults
- ☐ Low evaporator refrigerant temperature

The display also provides reports that are organized into six groupings: Evaporator, Condenser, Compressor, Motor, Purge, and the ASHRAE Chiller Log. Each report contains data that is accessed by scrolling through the menu items.

Each grouping will have a heading which describes the type of data in that grouping. This data includes:

- ☐ Phase currents
- ☐ Last 10 diagnostics
- ☐ Phase voltages
- ☐ Current limit setpoint
- ☐ Purge suction temperature
- ☐ Oil temperature and pressures
- ☐ Motor winding temperatures
- ☐ Current chiller operating mode
- ☐ All water temperatures and setpoints
- ☐ Condenser liquid refrigerant temperature
- ☐ Compressor starts and hours running
- ☐ Saturated refrigerant temperatures and pressures
- ☐ Control source (i.e., local panel, external source, remote BAS)

The controller is capable of receiving signals from a variety of control sources (which are not mutually exclusive—i.e., multiple control sources can coexist simultaneously) and of being programmed at the keypad as to which control source has priority. Control sources can be:

- ☐ The local operator interface
- ☐ Generic BAS (interface optional, control source not supplied by chiller manufacturer) The control source with priority will then determine the active setpoints via the signal that is sent to the control panel.

## 2.9 ISOLATION PADS

Isolation pads are supplied with each chiller for placement under all support points. They are constructed of molded neoprene.

## 2.10 STARTER

The starter type is a Unit Mount Primary Reactor Starter Factory mounted and completely pre-wired to compressor motor and control panel. The starter is a reduced voltage starter and therefore drew 65% of locker rotor amps, and the motor acceleration time is 3-8 seconds typically.

## PART 3 - EXECUTION

### 3.1 INSTALLATION

- A. All equipment and components shall be installed in a manner which shall ensure the proper operation as recommended by the manufacturer's installation instructions.
- B. Provide for connection to electrical service.
- C. Provide for connection of electrical wiring between starter and chiller control panel, oil pump to emergency power and separately fused circuit.
- D. Align chiller on concrete foundations, sole plates, and sub-bases. Level, grout, and bolt in place.
- E. Provide evaporator connections to chilled water piping. F.  
Furnish and install necessary auxiliary water piping for oil cooling units.
- G. Insulate evaporator and cold surfaces.
- H. Provide condenser connection to condenser water piping. I.  
Arrange piping for easy dismantling to permit tube cleaning.
- J. Provide piping from Pressure relief device to outdoors. Size as recommended by manufacturer.

### 3.2 TESTS AND START-UP

The following actions shall be supervised by the unit manufacturer's representative:

- A. Charging
- B. Start-up
- C. Field testing

### 3.3 DEHYDRATING

- A. All units shall be Site tested for leaks.

- B. Leak checking shall be made using a handle testing device.
- C. All units shall be dehydrated by producing a vacuum of 100 microns, using a high vacuum pump, with:
  - 1. The vacuum maintained for 4 hours.
  - 2. The pump stopped at the end of the 4 hours period.
  - 3. The vacuum maintained in the refrigeration unit for 24 hours without losing more than 50 microns.
  - 4. The Contractor shall certify that the tests have been made and shall submit 3 copies of the report(s) for the approval of the Engineer.

#### 3.4 FIELD REFRIGERANT PIPING TESTS SHALL:

- A. Be held in accordance with any requirements of local authorities having jurisdiction and a safety code for mechanical refrigeration.
- B. Be held by the unit manufacturers and certified in respect to the completion of charging, start-up and field testing, and the unit being made ready for proper operation.

#### 3.5 SERVICE

Upon completion and acceptance of each installation the Contractor shall provide the services of a competent instructors to properly acquaint the Client's staff with all details so as to enable them to obtain maximum efficiency of plant operation. The prior written acceptance of each instruction day shall be sought from the Client.

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